

Supplementary Material for Hierarchical Convexity-Run Decomposition: Finite-Sample Chord-Lobe Recovery and Interpretable LC–MS Transfer

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This supplement contains optional scan and post-selection theory, complete secondary experimental results, and detailed proof edges. Numbered references to theorems, figures, and tables in the main article use the main-article numbering.

1 Template scans for chord lobes

These are conservative sup-norm certificates. They are not sharp detection boundaries when location and scale are unknown. Modern multiscale shape tests and scan statistics provide the relevant calibration and minimax comparison targets [7, 1, 20].

The complete HCRD shapes also admit a sharper finite-dictionary statement. Let P project onto $\text{span}\{\mathbf{1}, x\}$ and let $M \geq 2$ and $v_1, \dots, v_M \in \text{range}(I - P)$ be unit-norm, oriented lobe templates. They may be fixed in advance or extracted from an independent guide replicate; in the latter case all statements below are conditional on that guide. Write $\rho_m = \max_{k \neq m} \langle v_k, v_m \rangle < 1$.

Theorem 1 (Lobe-dictionary detection, localisation, and lower bound). *For $Y = g + \epsilon$, $g \in \text{range}(P)$ and $\epsilon \sim N(0, \sigma^2 I)$, the scan*

$$S = \max_{1 \leq m \leq M} \frac{\langle v_m, Y \rangle}{\sigma} \tag{1}$$

has level at most α at threshold $t_\alpha = \sqrt{2 \log(M/\alpha)}$. Under $Y = g + \sigma \mu v_m + \epsilon$, its miss probability is at most β whenever

$$\mu \geq t_\alpha + \sqrt{2 \log(1/\beta)}. \tag{2}$$

Moreover, the maximiser of the scan equals m with probability at least $1 - \delta$ if

$$\mu(1 - \rho_m) > 2\sqrt{2 \log(2M/\delta)}. \tag{3}$$

Conversely, if the v_m are orthonormal and $\alpha + \beta < 1$, no level- α test has power greater than $1 - \beta$ uniformly over all M alternatives when

$$e^{\mu^2} - 1 \leq 4M(1 - \alpha - \beta)^2. \tag{4}$$

Under a uniform unknown index, every index estimator also has mean error at least $1 - (\mu^2 + \log 2)/\log M$. Thus both detection and localisation have critical standardised lobe norm of order $\sqrt{\log M}$. Over the stated orthonormal subclass, the scan therefore matches the lower-bound order $\sqrt{\log M}$, with an explicit constant gap; this is not a minimax claim for every correlated HCRD dictionary.

Proof. Under the null each scan coordinate is standard Gaussian, possibly correlated, so a Gaussian tail bound and union bound give $\Pr(S > t_\alpha) \leq M e^{-t_\alpha^2/2} = \alpha$. Under the stated alternative, the true coordinate is $N(\mu, 1)$, which proves the miss bound. With probability at least $1 - \delta$, all M centred coordinates have magnitude at most $u = \sqrt{2 \log(2M/\delta)}$. The true coordinate then exceeds every competitor when $\mu - u > \mu \rho_m + u$.

For the lower bound, restrict the affine nuisance to $g = 0$ and mix the M orthogonal alternatives uniformly. Its chi-squared divergence from the null is $(e^{\mu^2} - 1)/M$, hence total variation is at most one half of its square root. The displayed condition makes average power at most $1 - \beta$, so at least one alternative has no greater power. Finally the pairwise Gaussian Kullback–Leibler divergence is μ^2 ; Fano’s inequality [4] gives the localisation claim. $\square$

The theorem concerns fixed or independently selected residual shapes. It does not make the discontinuous same-replicate knot selector innocuous, and its orthogonal lower-bound subclass is a benchmark rather than a claim that all HCRD lobes are mutually orthogonal.

The finite dictionary is not the mathematical end point: location, scale, and asymmetry are naturally continuous. Let Θ be compact, let w_θ be raw lobe samples, and suppose $\inf_\theta \|(I - P)w_\theta\|_2 > 0$. Define

$$v_\theta = \frac{(I - P)w_\theta}{\|(I - P)w_\theta\|_2}, \quad d(\theta, \vartheta) = \|v_\theta - v_\vartheta\|_2, \quad (5)$$

and let $N(\epsilon)$ be the covering number in d . Write $J(\Theta) = \int_0^2 \sqrt{\log N(\epsilon)} d\epsilon$.

Theorem 2 (Continuous chord-lobe scan). *Assume $\theta \mapsto v_\theta$ is continuous, $J(\Theta) < \infty$, and the family is fixed or selected from an independent guide. For $Y = g + e$, $g \in \text{range}(P)$ and $e \sim N(0, \sigma^2 I)$, the continuous scan*

$$S_c = \sup_{\theta \in \Theta} \frac{\langle v_\theta, Y \rangle}{\sigma} \quad (6)$$

has level at most α at the safe threshold

$$t_\alpha = 24J(\Theta) + \sqrt{2 \log(1/\alpha)}. \quad (7)$$

Under $Y = g + \sigma \mu v_{\theta_0} + e$, its miss probability is at most β whenever $\mu \geq t_\alpha + \Phi^{-1}(1 - \beta)$.

If $G_\eta \subset \Theta$ is a finite η -net in d , scanning its M templates at $\sqrt{2 \log(M/\alpha)}$ still has level at most α . Every continuous alternative has a net template with correlation at least $1 - \eta^2/2$; hence the sieve has miss probability at most β whenever

$$\mu \geq \frac{\sqrt{2 \log(M/\alpha)} + \Phi^{-1}(1 - \beta)}{1 - \eta^2/2}, \quad \eta < \sqrt{2}. \quad (8)$$

Proof. Affine residualisation centres the Gaussian process $Z_\theta = \langle v_\theta, e \rangle / \sigma$; its canonical metric is exactly d and every marginal variance is one. Dudley’s entropy inequality bounds $\mathbb{E} \sup_\theta Z_\theta$ by $24J(\Theta)$ [6]. Gaussian concentration about this mean gives the stated tail [22]. At the true template the alternative score is $N(\mu, 1)$, proving the power statement. For the sieve, the finite union bound gives level and the unit-vector identity $\langle u, v \rangle = 1 - \|u - v\|_2^2/2$ gives the approximation factor. $\square$

Theorem 3 (Rank-aware projected-norm control for continuous scans). *Let every v_θ be a unit vector in a fixed or independent-guide q -dimensional subspace $U \subseteq \text{range}(I - P)$. If $Q_q(1 - \alpha)$ is the $(1 - \alpha)$ quantile of χ_q^2 , set*

$$t_{q,\alpha} = \sqrt{Q_q(1 - \alpha)}. \quad (9)$$

Then the same continuous scan has level at most α at $t_{q,\alpha}$. Its miss probability under $g + \sigma \mu v_{\theta_0}$ is at most β whenever

$$\mu \geq t_{q,\alpha} + \Phi^{-1}(1 - \beta). \quad (10)$$

For a parameter metric d_Θ , define

$$\Delta_{\theta_0}(r) = \inf_{d_\Theta(\theta, \theta_0) \geq r} \{1 - \langle v_\theta, v_{\theta_0} \rangle\}. \quad (11)$$

If this modulus is positive, every measurable scan maximiser lies within r of θ_0 with probability at least $1 - \delta$ whenever

$$\mu \Delta_{\theta_0}(r) > 2t_{q,\delta}. \quad (12)$$

A closed-form, dependency-free replacement for the projected-norm quantile is

$$\bar{t}_{q,\alpha} = \sqrt{q + 2\sqrt{q \log(1/\alpha)} + 2 \log(1/\alpha)}. \quad (13)$$

Proof. Under the affine null, $S_c \leq \|\Pi_U e / \sigma\|_2$ by Cauchy–Schwarz, simultaneously over the uncountable unit-template family. Because U is fixed relative to the scoring noise, the squared projected norm is χ_q^2 . Thus the χ_q quantile exactly calibrates the dominating projected norm and gives scan size at most α ; it is the scan’s exact null quantile only when the template family attains the relevant unit-sphere directions. The true-template coordinate is $N(\mu, 1)$, giving the stated sufficient power condition. On the projected-norm event at level δ , every signed noise inner product is bounded by $t_{q,\delta}$; the mean score gap outside radius r is at least $\mu \Delta_{\theta_0}(r)$, proving localisation. The last display follows from the Laurent–Massart chi-squared tail bound [14]. $\square$

No entropy or discretisation is needed for this conservative envelope in Theorem 3; compact continuity supplies measurability. For affine residualisation on a strictly increasing grid, $U = \text{range}(I - P)$ has $q = n - 2$. Selecting U from the scoring noise, using unnormalised templates, or replacing σ by a same-sample estimate invalidates the displayed pivot.

The entropy input can itself be certified analytically. If $\Theta = \prod_{j=1}^d [a_j, b_j]$ and $\|v_\theta - v_\vartheta\|_2 \leq L\|\theta - \vartheta\|_2$, coordinate grids give

$$N(\epsilon) \leq \prod_{j=1}^d \left(1 + \frac{L\sqrt{d}(b_j - a_j)}{\epsilon}\right), \quad (14)$$

which has an explicit finite entropy integral.

Corollary 1 (Interior triangular-family certificate). *Let $x_0 < \dots < x_{n-1}$ have maximum gap Δ . Parameterise a unit-height triangular lobe by centre c , support width s , and apex fraction a on a box whose supports lie strictly inside (x_0, x_{n-1}) . With*

$$h = s_- \min\{a_-, 1 - a_+\}, \quad \gamma = 1 - \frac{\Delta}{2h} > 0, \quad B = \sqrt{\frac{7}{2} + \max\{|a_- - 1/2|, |a_+ - 1/2|\}^2 + s_+^2}, \quad (15)$$

the residuals and normalized template map satisfy

$$\inf_{\theta} \|(I - P)w_\theta\|_2 \geq \frac{\gamma}{\sqrt{2}}, \quad \|v_\theta - v_\vartheta\|_2 \leq \frac{4\sqrt{n}B}{h\gamma} \|\theta - \vartheta\|_2. \quad (16)$$

Thus Theorem 2 has a fully analytic certificate for this declared location–width–asymmetry family.

Proof. A grid point lies within $\Delta/2$ of every apex, so the sampled height is at least γ . Contrast that sample using a vector supported on it and the two zero domain-endpoint samples and orthogonal to 1 and x to obtain the residual lower bound. On each triangle branch, derivatives with respect to the left, apex, and right knots are bounded by $\sqrt{2}/h$. Combining their parameter Jacobian bound B , summing over samples, projecting nonexpansively, and applying the $2/m$ normalization inequality with $m = \gamma/\sqrt{2}$ gives the stated Lipschitz constant. $\square$

The constants are deliberately safe. Boundary-touching supports, general polygonal lobes, estimated or dependent noise, and sharper generic chaining remain outside this corollary.

2 Replicate-split inference for selected structures

Let a guide G be independent of a scoring replicate $Y = f + e$, $e \sim N(0, \sigma^2 I)$. HCRD may select any finite family of intervals and shapes from G . On an active interval I_s , let c_s be the trapezoidal coefficient vector for signed area relative to the endpoint chord; it annihilates every affine vector. Alternatively let $q_s(G)$ be the complete guide detail and $v_s = (I - P_s)q_s(G)$, where P_s projects onto $\text{span}\{\mathbf{1}, x\}$.

Theorem 4 (Exact post-selection area and shape pivots). *Conditionally on G , every nondegenerate selected structure satisfies*

$$\frac{c_s^T Y_{I_s}}{\sigma \|c_s\|_2} \sim N\left(\frac{c_s^T f_{I_s}}{\sigma \|c_s\|_2}, 1\right), \quad \frac{v_s^T Y_{I_s}}{\sigma \|v_s\|_2} \sim N\left(\frac{v_s^T f_{I_s}}{\sigma \|v_s\|_2}, 1\right). \quad (17)$$

Two-sided Gaussian p -values followed by Holm correction control conditional and unconditional family-wise error over all selected, overlapping structures. Under $f_{I_s} = a + bx + a_s v_s$, the matched-shape noncentrality is $a_s \|v_s\|_2 / \sigma$.

Proof. Conditionally on the independent guide, c_s and v_s are fixed. Fixed linear functionals of the Gaussian scoring noise have the displayed laws, and both coefficient vectors annihilate affine means. Holm control requires valid marginal p -values but no independence among the overlapping tests. $\square$

Reusing the scoring noise to select its largest structure invalidates these pivots; with two independent null candidates, selecting the smaller naive p -value already inflates rejection probability to $1 - (1 - \alpha)^2$. Independent replicates are therefore a substantive assumption, not a notation for adjacent samples in one autocorrelated series.

Theorem 5 (Buffered single-stream post-selection pivots). *Let $Y_t = f_t + e_t$, where e is a mean-zero Gaussian m -dependent process. Partition the stream into contiguous scoring blocks B_k and let guide σ -field $\mathcal{G}_k$ use only indices at distance greater than m from B_k . Suppose a finite family of contrasts c_{ks} , supported on B_k , is $\mathcal{G}_k$ -measurable and the positive block covariance Σ_{B_k} is known. Under $H_{ks} : c_{ks}^T f_{B_k} = 0$,*

$$Z_{ks} = \frac{c_{ks}^T Y_{B_k}}{\sqrt{c_{ks}^T \Sigma_{B_k} c_{ks}}} \bigg| \mathcal{G}_k \sim N(0, 1). \quad (18)$$

Consequently two-sided Gaussian p -values remain valid after guide-side HCRD selection, and Holm correction over every fold and selected structure controls strong family-wise error without assuming cross-fold independence.

Proof. m -dependence and the buffer make e_{B_k} independent of $\mathcal{G}_k$. Conditional on that guide, each contrast is fixed; Gaussianity and the full block covariance give the displayed law. Holm requires valid marginal p -values, not mutual independence. $\square$

Every index can be scored once: only the m neighbours of a scoring block are omitted from that fold's guide, not from their own scoring folds. The theorem is exact for known-covariance Gaussian finite-memory noise. An AR(1) process is not m -dependent for any finite m ; estimated covariance, data-selected lag, and general mixing therefore require additional error control.

Theorem 6 (Nested subtree gatekeeping). *Let T be a fixed or independent-guide finite rooted tree and associate to each node v the intersection null H_v that there is no signal anywhere in the subtree rooted at v . Let p_v be marginally valid for H_v under arbitrary dependence. The tree and*

all local level allocations are deterministic or measurable with respect to the independent guide. Assign $a_{\text{root}} = \alpha$ and local levels satisfying

$$\sum_{u \in \text{child}(v)} a_u \leq a_v. \quad (19)$$

Test the root at a_{root} and a non-root node at a_v only after every ancestor rejects. Then the probability of rejecting any true null is at most α .

Proof. The minimal true nodes form an antichain, and rejection of any true descendant requires rejection of one such boundary node. Child-budget conservation implies inductively that the levels assigned to any antichain sum to at most α . A union bound over the minimal true nodes completes the proof. $\square$

Allocating a parent’s budget to children in proportion to descendant leaf counts conserves the full budget and retains level α along a one-child chain. This may improve power for coherent nested alternatives but does not uniformly dominate Holm for isolated leaves. Geometric containment alone is insufficient: the node nulls must be subtree intersection nulls and a same-scoring-noise tree does not automatically have valid node p-values.

3 Additional experiments and audits

3.1 Synthetic, comparator, and runtime experiments

Table 1: Mean baseline MSE over 100 preregistered trials. Oracle labels indicate per-signal ground-truth tuning.

Method	Exact, $\sigma = 0$	Exact, $\sigma = 0.1$	Variable, $\sigma = 0.1$
Centred HCRD	0	0.787	0.887
Thresholded HCRD	3.78e-32	0.00778	0.312
Adaptive guided HCRD	0.0124	0.0385	0.0856
Gaussian oracle	0.0433	0.0382	0.0632
Moving-average oracle	0.0516	0.0486	0.0743

As a computational audit rather than a proof, we also simulated 50,000 null and alternative draws for a fixed 24-template compact-lobe dictionary in 257 samples. Its maximum coherence was 0.0376. At the conservative theorem threshold for $\alpha = 0.05$, empirical false rejection was 0.00524 (Wilson interval $[0.00464, 0.00591]$). At the sufficient power norm, miss probability was 0.0363 versus the guaranteed bound 0.20; at the localisation norm, no error was observed (upper Wilson limit 7.68×10^{-5}) versus the bound 0.10. These checks expose conservatism and implementation errors; they do not replace Theorem 1.

We separately audited Theorem 2 on 129 samples and a 595-template centre–width–asymmetry sieve. In 50,000 null draws the finite threshold 4.332 was exceeded 22 times: empirical FWER 0.00044 with 95% Wilson interval $[0.000276, 0.000666]$. Its true-template Gaussian sufficient norm 5.1739 for $\beta = 0.2$ gave empirical power 0.8773 in 10,000 draws. For the complete continuum, $q = 127$: the projected-norm envelope threshold 12.4218 was exceeded in 2537/50,000 direct projected-norm draws, giving 0.05074 ($[0.04883, 0.05270]$), while the closed-form Laurent–Massart threshold 13.1150 was exceeded with probability 0.00486. The resulting conservative 80%-power sufficient norm is 13.2634. This replaces the earlier Dudley sufficient norm 285.1103 without treating the numerical grid as the continuum. These draws audit the dominating norm, not the generally smaller scan null. The finite sieve remains sharper because it declares only 595 templates rather than the full residual subspace.

On 100 fresh noiseless exact-class trials, centred HCRD recovered the latent baseline to numerical zero. It had lower MSE in every trial than the historical legacy rule and the oracle-tuned Gaussian, moving-average, and Fourier comparators. All corresponding paired bootstrap intervals were below zero and Holm-adjusted sign-test values were below 1.6×10^{-29} . Thresholded HCRD and an endpoint-only RDP baseline were also exact within 10^{-12} and are reported as ties, not losses. That RDP solution uses only the two domain endpoints: it recovers the global affine baseline but not the $J - 1$ lobe boundaries, so it does not tie the boundary-recovery endpoint in Figure 2. On the exploratory variable-amplitude, piecewise-affine class without noise, centred HCRD had mean MSE 0.0179, versus 0.0645 for the Gaussian oracle, but the raw method failed under sample-level white noise. Adaptive guidance improved knot accuracy substantially; its pilot-calibrated scale is not yet supported by a distribution-free theorem.

An external-method experiment then used 50 further exact-class trials. On the noiseless condition, centred HCRD had lower baseline MSE in all 50 trials than the final EMD residue, the lowest-frequency VMD mode, and per-signal oracle-tuned ℓ_1 trend filtering. The respective mean competitor MSEs were 0.0108, 0.511, and 0.0425, versus zero for HCRD; every paired bootstrap interval was below zero and the Holm-adjusted sign-test value was 1.07×10^{-14} . A supplementary direct comparison on the same 50 fixed draws used the final ITD baseline returned by PySDKit 0.4.54. ITD mean MSE was 0.1473; the HCRD–ITD paired difference was -0.1473 ($[-0.1871, -0.1085]$), with 50/50 HCRD wins and Holm-adjusted exact sign $p = 1.24 \times 10^{-14}$. EMD, VMD, and ITD settings follow their standard algorithms [11, 5, 9]; oracle tuning deliberately favours the smoothing comparators. This confirmation is specific to the chord-lobe class, rather than evidence of global dominance over ITD’s time–frequency use cases. Although DPT is an exact nonlinear pulse decomposition, its native components are constant pulses and do not define a unique final chord baseline at a matched scale; it is therefore included in the structural comparison rather than assigned an artificial baseline-MSE endpoint.

After that result, two fresh 100-trial noisy confirmations were locked on new seeds before outcome inspection for the plug-in MAD-thresholded variant. Table 2 reports their means. At $\sigma = 0.03$, its win rates against EMD, VMD, ℓ_1 , and Gaussian were respectively 0.86, 0.99, 0.99, and 0.98; at $\sigma = 0.10$ they were 0.81, 0.93, 0.96, and 0.74. All paired bootstrap intervals remained below zero. The largest Holm-adjusted sign-test value was 1.67×10^{-6} . This finite-sample confirmation concerns the specified generator and plug-in estimator; Theorem 9 itself assumes known or externally estimated noise scale.

Table 2: Mean baseline MSE for external comparisons. The noiseless column is E1 (50 trials); noisy columns are fresh C2 confirmations (100 trials each).

Method	$\sigma = 0$	$\sigma = 0.03$	$\sigma = 0.10$
Centred HCRD	0	0.762	0.789
Thresholded HCRD	2.92×10^{-32}	0.00356	0.0195
EMD residue	0.0108	0.0288	0.0531
VMD low mode	0.511	0.0344	0.0396
ℓ_1 trend oracle	0.0425	0.0419	0.0443
Gaussian oracle	0.0378	0.0336	0.0383

Two additional external decompositions were then tested under independently frozen, fresh-seed protocols. Improved complete ensemble EMD with adaptive noise (CEEMDAN) [3] used 20 noise realizations and an oracle selected, separately for every signal, the cumulative one-to-four slowest components that minimized latent-baseline MSE. Fast Iterative Filtering used the author-linked Python implementation with its published defaults [2]. Its oracle used the analogous cumulative slow-tail selection. Both oracles use unavailable ground truth and therefore favour the comparators.

Table 3 reports the results. In E2, HCRD won 50/50, 48/50, and 35/50 paired trials at noise standard deviations 0, 0.03, and 0.10; the largest Holm-adjusted sign-test value was 0.00660, and all three 95% bootstrap intervals for HCRD minus CEEMDAN lay below zero. In E3 the corresponding wins over oracle Iterative Filtering were 50/50, 50/50, and 49/50, with largest adjusted value 9.06×10^{-14} and again all intervals below zero. These confirmations strengthen only the stated chord-lobe baseline task: CEEMDAN and Iterative Filtering target oscillatory mode decompositions rather than endpoint-chord recovery.

Table 3: Fresh modern-decomposition confirmations (50 signals per condition). The HCRD reference is centred at $\sigma = 0$ and MAD-thresholded otherwise. E2 and E3 use independent seeds and must be read as paired rows within study.

Study	Method	$\sigma = 0$	$\sigma = 0.03$	$\sigma = 0.10$
E2	HCRD reference	0	0.000400	0.00636
E2	CEEMDAN oracle slow tail	0.0135	0.0137	0.0312
E3	HCRD reference	0	0.000398	0.0124
E3	Iterative Filtering oracle tail	0.135	0.156	0.143

Finally, an exploratory Case Western Reserve University bearing study used 12 kHz drive-end recordings for four states and four motor loads [21]. Complete recordings, rather than randomly mixed windows, defined leave-one-load-out folds. Raw summary features already reached mean balanced accuracy 0.982; Gaussian, wavelet, EMD, and raw HCRD representations all reached 1.000. Thus the data exhibit a ceiling effect and provide no evidence that HCRD is superior. Guided HCRD was worse at 0.945, including 0.781 on the 3-hp fold. The experiment therefore identifies a ceiling regime in which HCRD provides no classification benefit.

Runtime protocol P1 processed the same 384 CWRU windows five times per mode in randomized order after a whole-system load gate; compiled-library thread counts were fixed to one, and pool startup and serialization were timed. Every one of the 35 feature matrices was bitwise identical to the saved R1 matrix. On an Intel Core Ultra 7 155H, serial median latency was 10.279 s. Two, four, eight, and sixteen worker processes achieved respectively 1.76, 2.16, 2.47, and 2.00 times speedup; four and eight Python threads achieved only 0.55 and 0.60 times serial performance because the reference knot walk is GIL-bound. The complete machine-specific scaling results are provided in the repository. Levels within one signal remain sequential, whereas independent windows, records, and channels are exactly parallel.

We implemented Corollary 1 by storing only nested knot sets and materializing dense arrays on demand. Protocol P2R required five consecutive one-second whole-system samples, each at most 20%, before every trial. Every timed hierarchy had the same canonical knot digest, and materialized features for all 384 windows were bitwise identical to R1. Sparse serial construction reduced the median from 14.461 to 1.217 s, an 11.89-fold representation-level speedup; four processes further reduced it to 0.976 s (1.25-fold versus sparse serial). Eight processes did not amortize startup and serialization on this short batch.

Protocol P3 therefore froze a tenfold larger 3840-window throughput batch under the same per-trial load gate. Table 4 shows monotone process scaling to a 1.96-fold end-to-end speedup with eight workers. This supports exact independent-signal parallelism, but not ideal linear scaling; all timings include pool construction, serialization, output collection, and hierarchy allocation.

Table 4: Strictly load-gated sparse runtime results. Speedup is relative to dense serial in P2R and sparse serial in P3.

Protocol	Representation/backend	Workers	Signals	Median (s)	Speedup
P2R	dense serial	1	384	14.461	1.00
P2R	sparse serial	1	384	1.217	11.89
P2R	sparse process	4	384	0.976	14.82
P3	sparse serial	1	3840	13.682	1.00
P3	sparse process	2	3840	11.911	1.15
P3	sparse process	4	3840	8.669	1.58
P3	sparse process	8	3840	6.963	1.96

A separate new-seed confirmation assessed the quadratic proximal guide with a single pilot-selected but then frozen value $\lambda = 3$ on the variable-amplitude chord-lobe generator. Thirty independent latent signals, each with ten noise repetitions, were used at every noise level; inference averaged repetitions within latent signal. Table 5 shows a morphology advantage. Against the adaptive Gaussian guide, paired F1 differences were 0.274, 0.244, and 0.191 for $\sigma = 0.01, 0.03, 0.05$, with all 95% bootstrap intervals strictly positive; at $\sigma = 0.10$ the difference was 0.011 with interval $[-0.015, 0.039]$, an explicit tie. Quadratic guidance also exceeded raw, thresholded, and fixed-Gaussian HCRD at every noise level after Holm correction. The maximum sampled guide perturbation ratio was 0.366, consistent with Theorem 7; after hard HCRD it reached 2.543, confirming that the global theorem must not be transferred to the knot output.

Table 5: Frozen S2 confirmation. F1 matches latent chord knots within one sample. MSE is against the latent baseline.

σ	Quad. F1	Adaptive F1	Fixed-Gauss. F1	Quad. MSE	Adaptive MSE
0.01	0.716	0.442	0.410	0.0730	0.0722
0.03	0.720	0.476	0.399	0.0732	0.0863
0.05	0.722	0.531	0.338	0.0732	0.0922
0.10	0.720	0.709	0.230	0.0727	0.0964

3.2 E2 qscore implementation sensitivity

The main E2 comparison used a global-window independent reimplement of the published qscore. We repeated both transfer directions after changing the minimum-point rule and the across-file summaries, always comparing HCRD-8 plus a given qscore variant with that same variant alone. Table 6 reports the complete paired analysis.

Table 6: qscore implementation sensitivity. Entries are HCRD-8 plus the indicated qscore summary minus that same qscore alone in target AP, with paired target-feature 95% bootstrap intervals.

Variant	Width	Falkor $\rightarrow$ MESOSCOPE	MESOSCOPE $\rightarrow$ Falkor
Current median, $m \geq 8$	2	0.1181 [0.0628, 0.1769]	0.1005 [0.0554, 0.1457]
Median, $m \geq 5$	2	0.1244 [0.0688, 0.1832]	0.0987 [0.0538, 0.1432]
Author-like five-summary, $m \geq 5$	5	0.1517 [0.0813, 0.2184]	0.0765 [0.0324, 0.1211]
Multisummary seven-variable, $m \geq 8$	7	0.0847 [0.0320, 0.1407]	0.0798 [0.0347, 0.1262]

3.3 Expert-labelled chromatographic lobes

We evaluate HCRD for LC-MS peak-shape curation. A genuine extracted-ion chromatogram (EIC) is approximately a compact, predominantly one-sign lobe over a local baseline, whereas shoulders, split peaks, baseline fragments, and noise violate that structure. The public data of Müller et al. [17] contain 255,000 EICs: 5000 candidate ions evaluated in each of 51 real samples by mass-spectrometry experts. Before reading the arrays or classification table, we froze a double group holdout. Sample identifiers and candidate-ion identifiers were hashed independently; a case was admitted only when both hashes assigned it to the same train, validation, or confirmation block. Confirmation thus shared neither samples nor candidate ions with model development.

Every representation received both time windows shown to the experts. The same fixed histogram-gradient-boosted classifier was trained on robustly resampled raw EICs, a conventional multiscale morphology bank, raw plus one HCRD level, raw plus the complete eight-level HCRD hierarchy, HCRD scalar geometry without waveform channels, or polygon-area/energy summaries alone. Validation selected the conventional domain bank before the confirmation block was unlocked. Primary inference used a 10,000-replicate two-way cluster bootstrap over both held-out samples and held-out candidate ions.

Table 7 shows the locked result. Eight-level HCRD achieved the largest confirmation AP, 0.4092 versus 0.3820 for the frozen comparator, and exceeded the one-level representation by 0.0321. The primary difference was 0.0272, but its cluster interval $[-0.0566, 0.1010]$ crossed zero because the double holdout left only eight independent confirmation samples. The pre-specified success rule therefore failed. The ablations are nevertheless informative: area alone was weakest, and scalar geometry without the signed detail channels was also below the conventional comparator. The useful object is the complete multilevel decomposition in combination with the raw shape; polygon mass is auxiliary rather than the method.

Table 7: Locked E1 expert-labelled LC-MS confirmation (2692 unambiguous EICs, 297 peaks; eight sample and 706 candidate-ion groups). AP is the primary threshold-free endpoint.

Representation	AP	ROC AUC	MCC	Balanced acc.
Raw EIC	0.3626	0.8714	0.4150	0.7782
Conventional domain bank	0.3820	0.8801	0.4379	0.7866
Raw + HCRD level 1	0.3771	0.8696	0.4026	0.7594
Raw + HCRD levels 1–8	0.4092	0.8844	0.4418	0.8133
HCRD scalar geometry	0.3345	0.8415	0.3479	0.7409
Polygon area/energy only	0.3151	0.8254	0.3369	0.7451

3.4 Application-boundary and cross-domain studies

Two additional tests evaluated the application boundary. LCMS-E3 used the public TargetedMSQC CSF panel and leave-one-run-out transfer across four runs [8]. This is a deliberately morphology-saturated setting: its 52 official variables already encode jaggedness, symmetry, modality, width, elution shift, transition ratios, and light/heavy consistency. On 576 eligible labelled transition pairs, QC52 AP was 0.9247 versus 0.8862 after adding the complete HCRD-8 bank. The difference was -0.0384 with a two-way run/peptide cluster interval $[-0.1455, 0.0505]$, and HCRD won only one of four held-out runs. HCRD-8 did exceed HCRD-1, but the pre-specified test failed. A later compact-geometry search reached AP 0.9372 and is reported only as development because it used these outcomes.

LCMS-E4 tested baseline recovery on 60 low-affine-deviation TIC windows from 20 raw NeatMS Dataset 1 mzML files [10]. Single and nested positive lobes were injected after the protocol and baselines were scored against their known real-background target. For nested lobes, HCRD-8 MSE was 0.1369, versus 0.1141 for per-case oracle asymmetric least squares and 0.1270

for oracle morphological opening. It nevertheless beat HCRD-1, Gaussian, Savitzky–Golay, and wavelet comparators on all 60 cases; the HCRD-8–HCRD-1 mean difference was -0.3942 with simultaneous interval $[-0.4082, -0.3801]$. A post-protocol mixed-polarity analysis remained worse than signed AsLS and symmetric morphology. Table 8 summarizes these task-specific boundaries.

Table 8: LC–MS application-boundary tests. Lower MSE is better in E4.

Test	HCRD	Comparator	Boundary
E3 TargetedMSQC	AP 0.8862	QC52 AP 0.9247	morphology-saturated features
E4 real background	MSE 0.1369	AsLS MSE 0.1141	sign-aware lower envelope

3.5 Cross-domain boundary searches

The hierarchy was also tested as a learned pulse-event representation and as a training-free transient score. Table 9 summarizes the results; full protocols, folds, plots, bootstrap intervals, and ablations are provided in the repository.

Table 9: Cross-domain boundary searches. Each row reports the corresponding held-out or cross-fitted comparison; none is pooled with the LC–MS evidence.

Test	Main result	Interpretation
CWRU bearings [21]	Raw features 0.982 balanced accuracy; several transforms, including HCRD, tied at 1.000	Ceiling effect; no HCRD superiority
QTDB QRS [13, 12]	Quadratic HCRD 22.74 ms; ecgpuwave 13.94/15.75 ms	Stable guide improves raw HCRD, but the specialist delineator is better
PPG-DaLiA [19]	Geometry F1 0.7945, mass-only 0.7659, tuned find_peaks 0.8013	Full hierarchy carries information beyond area, without detector superiority
TSB-AD/Yahoo [16, 18]	HCRD 0.5285 AUC–PR versus NORMA 0.3383; CNN 0.9052	Positive frozen replication, but only within the synthetic Yahoo family
Real WSD/AIOps [15]	HCRD 0.3756 versus POLY 0.4115; hybrid 0.3675 versus raw spectral residual 0.3850	No independent real-KPI confirmation

For pulse data, the full multilevel geometry significantly exceeded the area/energy-only ablation, confirming that polygon mass is auxiliary rather than the decomposition. For temporal anomaly detection we used $S(t) = \max_{1 \leq \ell \leq 8} z_\ell(t)$, where z_ℓ is robust positive surprise in the levelwise area-density series. Its frozen Yahoo success did not transfer to real WSD or AIOps KPIs and therefore does not establish general anomaly-detection performance.

3.6 Stability and failure-mode audits

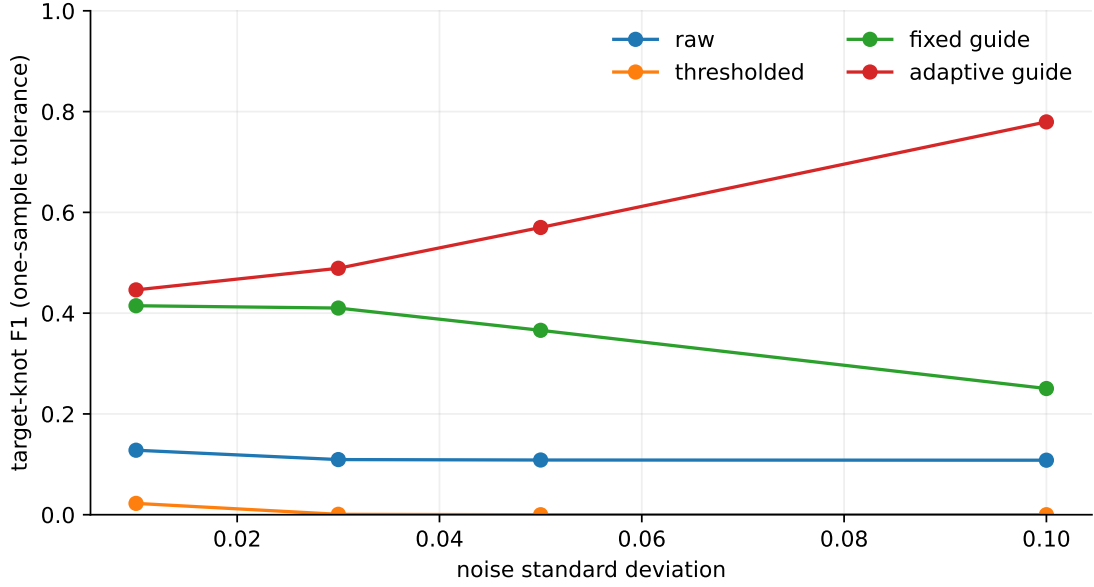


Figure 1: Knot accuracy under repeated Gaussian perturbations. Stability alone is insufficient: the thresholded method is stable because it often retains only endpoints, whereas adaptive guidance retains a compact set of meaningful knots.

Theorem 8 was audited on 1000 seeded perturbations, split equally between uniform and irregular grids. No numerical violation occurred; the maximum observed ratio $d_{SC}/(C_x\|e\|_\infty)$ was 0.9899. On the four-sample discontinuity sequence, hard-baseline amplification grew from 20.5 to approximately 2×10^8 as the curvature scale decreased from 10^{-1} to 10^{-8} , while the persistence distance remained exactly one half of its theorem bound up to floating-point error (Figure 2).

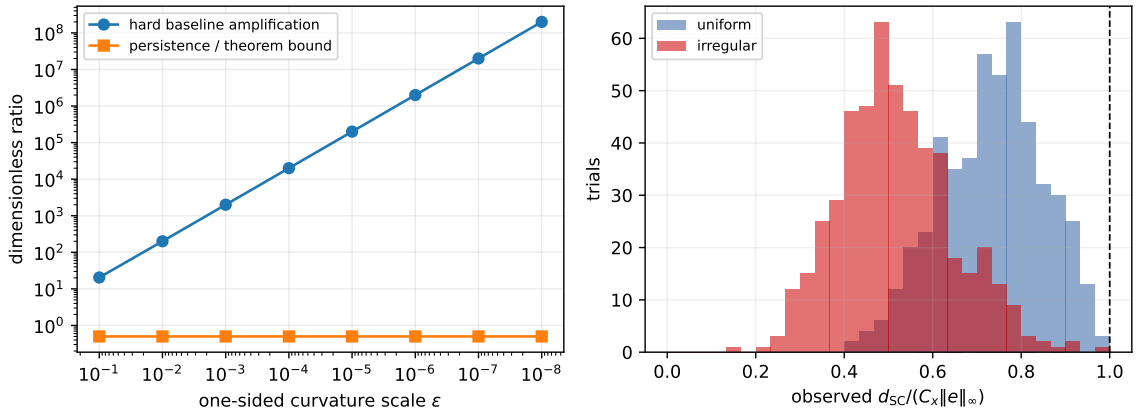


Figure 2: Hard-decision discontinuity versus signed-curvature persistence. Left: the hard baseline has unbounded amplification while the persistence distance remains within its global bound. Right: all 1000 uniform/irregular-grid checks remain below the theorem limit (dashed line).

A Boundary cases and detailed proof edges

This appendix expands the combinatorial and stability edges most vulnerable to off-by-one or hidden-decision assumptions. All indices below are local indices in the current eligible knot list; mapping them back to original sample indices preserves order and nesting.

Lemma 1 (Progress of a nonterminal centred interval). *Suppose the current interval starts at eligible index r and the centred walk closes it at t before the terminal endpoint. Then $t - r \geq 2$.*

Proof. Let $p \geq r + 1$ be the first active curvature after any insignificant entries. A nonterminal closure requires a later active curvature $q > p$ of the opposite sign. There are three possible selected locations. The first opposite point is $q \geq p + 1 \geq r + 2$. A single eligible zero between the two signs is selected at one index after the last active point, again at least $p + 1$. On a longer zero plateau, the last point on the departing side is eligible only when it is not the first active point; hence it is at least $p + 1$. Otherwise the arriving side is selected. The implementation's final spacing guard enforces the same lower bound in degenerate branches. Thus every nonterminal case has $t - r \geq 2$. $\square$

Lemma 2 (Halving recurrence). *If one level starts with $N \geq 1$ eligible intervals and produces N' chord intervals, then $N' \leq \lceil N/2 \rceil$.*

Proof. The produced intervals are consecutive and partition the N old intervals. By Lemma 1, every produced interval except possibly the last contains at least two old intervals. The last contains at least one. Hence $N \geq 2(N' - 1) + 1 = 2N' - 1$, so $N' \leq (N + 1)/2$ and integrality gives the claim. If the last interval also contains at least two, the stronger $N' \leq \lfloor N/2 \rfloor$ holds. $\square$

Detailed completion of Theorem 2 and Corollary 1. Every new knot is selected from the current eligible list, so the mapped knot sets are nested. Let $N_0 = n - 1$ and let N_j be the number of intervals after level j . Lemma 2 and the identity $\lceil \lceil a/2 \rceil / 2 \rceil = \lceil a/4 \rceil$ give inductively $N_j \leq \lceil N_0/2^j \rceil$. For $n \geq 3$, choosing $j = \lceil \log_2(n - 1) \rceil$ makes $N_j = 1$, so the endpoint chord has been recorded and the implementation stops. For $n = 2$, it records that chord in one level. This proves the stated $\max\{1, \lceil \log_2(n - 1) \rceil\}$ bound. If a sparse implementation spends $O(N_j)$ work at level j , the geometric recurrence gives $O(n)$ total work. More explicitly, for integer $N_0 \geq 1$, $\lceil N_0/2^j \rceil \leq (N_0 - 1)/2^j + 1$, so $\sum_{j=1}^J N_j < N_0 - 1 + J \leq 2N_0$. Adding one endpoint per stored knot set gives $\sum_{j=1}^J |K_j| < 2N_0 + J$, the stated storage bound. This is the complexity of the implemented knot-only representation; materializing every length- n level remains $O(n \log n)$ in the worst case. $\square$

Lemma 3 (Centred decision margin). *On a uniform grid of spacing h , define γ and η as in Theorem 5. If $\|e\|_\infty < \min\{\gamma h/4, \eta h/8\}$, every curvature sign and every adjacent opposite-sign magnitude comparison made by the centred walk is unchanged.*

Proof. The divided-curvature perturbation stencil is $(1, -2, 1)/h$, so every coordinate changes by at most $4\|e\|_\infty/h$. The first strict inequality prevents zero crossing. For two curvature coordinates a, b , the reverse triangle inequality gives

$$|(|a + \delta a| - |b + \delta b|) - (|a| - |b|)| \leq |\delta a| + |\delta b| \leq 8\|e\|_\infty/h.$$

The second strict inequality therefore preserves every comparison outcome. Once signs are fixed there are no zero-curvature branches, and the potential comparison pairs are the adjacent entries across sign-run boundaries. These data determine the same knot walk. $\square$

Detailed completion of Theorem 5. Lemma 3 fixes the knot set. At any sample between two fixed knots, the interpolation error is a convex combination of the two endpoint errors and is therefore at most $\|e\|_\infty$. This proves the baseline bound. The detail perturbation is $e - [T(y + e) - Ty]$,

so the triangle inequality gives the factor two. If $\eta = 0$, no positive centred-rule radius follows: the explicit $(1, 1, -1)$ counterexample in the main text changes the selected side while preserving every sign. $\square$

Lemma 4 (Essential class in the signed persistence distance). *For a nonempty connected finite path, the ordinary zero-dimensional superlevel diagram has exactly one essential class. Under a finite-cost bottleneck matching between two functions on that path, the essential classes match each other; the remaining matching restricts to their finite diagrams.*

Proof. At the bottom of the filtration the entire path is present and connected, so exactly one component survives indefinitely. A diagram point with an infinite death coordinate has infinite distance from the diagonal and from every finite point. Therefore any finite-cost matching pairs the two essential points. Their cost is the absolute difference of their birth heights, which are the two function maxima. Removing that pair leaves precisely a matching of the finite parts. $\square$

Detailed completion of Theorem 8. The i th divided-curvature row is

$$(h_{i-1}^{-1}, -(h_{i-1}^{-1} + h_i^{-1}), h_i^{-1}),$$

whose absolute row sum is $2(h_{i-1}^{-1} + h_i^{-1})$. Taking the maximum gives the exact induced ℓ_∞ norm C_x . Coordinatewise positive part is 1-Lipschitz, so both signed vertex functions differ by at most $C_x \|y - z\|_\infty$. Their piecewise-linear extensions over a common edge differ by a convex combination of endpoint differences and obey the same bound. Apply persistence-diagram stability to their negatives, which converts superlevel filtrations to sublevel filtrations without changing the sup norm. Lemma 4 identifies the essential and finite costs used in d_{SC} . Taking the maximum over signs proves the displayed distance bound. Finally, a finite bar (b, d) has diagonal cost $(b - d)/2$; if its lifetime is strictly greater than twice that bound, a bottleneck matching attaining the bound cannot send it to the diagonal. $\square$

Domain detail for Theorem 7. For $n \geq 3$, the divided-curvature matrix has rank $n - 2$: given any target curvature vector, choose an initial value and slope and integrate the target curvatures successively to construct a preimage. Thus D is surjective. Since a proper ϕ has nonempty domain, $\phi \circ D$ is proper; it is closed and convex by continuity and linearity of D . Adding $\|y - z\|_2^2/2$ makes the objective coercive and strongly convex, so its minimizer exists and is unique. The standard proximal theorem now applies to $g = \phi \circ D$. If $P = \text{prox}_g$ is firmly nonexpansive, then so is $I - P$: writing $d = y - z$ and $p = Py - Pz$, firm nonexpansiveness gives $\|p\|^2 \leq \langle p, d \rangle$, which directly implies

$$\|(I - P)y - (I - P)z\|^2 \leq \langle (I - P)y - (I - P)z, y - z \rangle.$$

Finally, $D\ell = 0$ gives $g(w + \ell) = g(w)$; translating the optimization variable establishes affine equivariance. The $n = 2$ case has the zero curvature map and is immediate. $\square$

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